

This guide provides step-by-step solutions for **Assignment 1 - Part B (Uneven Cash Flow)**, focusing on calculating the Present Value (PV) and Future Value (FV) of non-constant cash flow streams.

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## Question 1: Stellar Packaging Ltd. Investment

**Problem:** Calculate the PV and FV (at year 5) for the following cash flows at a 10% rate.

- **Year 1:** 30,000 | **Year 2:** 45,000 | **Year 3:** 35,000 | **Year 4:** 50,000 | **Year 5:** 40,000.

### a) Calculate the Present Value (PV)

To find the PV, we discount each individual cash flow back to Year 0 using the formula:  $PV = \frac{CF}{(1+r)^n}$ .

- **PV Year 1:**  $\$30,000 / (1.10)^1 = 27,272.73\$$
- **PV Year 2:**  $\$45,000 / (1.10)^2 = 37,190.08\$$
- **PV Year 3:**  $\$35,000 / (1.10)^3 = 26,296.02\$$
- **PV Year 4:**  $\$50,000 / (1.10)^4 = 34,150.67\$$
- **PV Year 5:**  $\$40,000 / (1.10)^5 = 24,836.85\$$
- **Total PV:**  $\$27,272.73 + 37,190.08 + 26,296.02 + 34,150.67 + 24,836.85 = \textbf{PKR}\ 149,746.35\$$

### b) Calculate the Future Value (FV) at Year 5

To find the FV, we compound each cash flow forward to Year 5:  $FV = CF \times (1+r)^{5-n}$ .

- **FV Year 1:**  $\$30,000 \times (1.10)^4 = 43,923\$$
- **FV Year 2:**  $\$45,000 \times (1.10)^3 = 59,895\$$
- **FV Year 3:**  $\$35,000 \times (1.10)^2 = 42,350\$$
- **FV Year 4:**  $\$50,000 \times (1.10)^1 = 55,000\$$
- **FV Year 5:**  $\$40,000 \times (1.10)^0 = 40,000\$$
- **Total FV:**  $\$43,923 + 59,895 + 42,350 + 55,000 + 40,000 = \textbf{PKR}\ 241,168\$$

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## Question 2: Project Comparison (Alpha, Beta, Gamma)

**Problem:** Compare three projects over 4 years at a 10% rate.

| Year | Alpha  | Beta   | Gamma  |
|------|--------|--------|--------|
| 1    | 40,000 | 35,000 | 50,000 |
| 2    | 30,000 | 45,000 | 30,000 |
| 3    | 25,000 | 40,000 | 25,000 |
| 4    | 20,000 | 55,000 | 25,000 |

#### a) Present Value (PV) Analysis

- **Project Alpha PV:**  $\$36,364 + 24,793 + 18,783 + 13,660 = \mathbf{\$93,600}$
- **Project Beta PV:**  $\$31,818 + 37,190 + 30,053 + 37,566 = \mathbf{\$136,627}$
- **Project Gamma PV:**  $\$45,455 + 24,793 + 18,783 + 17,075 = \mathbf{\$106,106}$

#### b) Future Value (FV) at Year 4

- **Project Alpha FV:**  $\$40k(1.1)^3 + 30k(1.1)^2 + 25k(1.1)^1 + 20k = \mathbf{\$137,040}$
- **Project Beta FV:**  $\$35k(1.1)^3 + 45k(1.1)^2 + 40k(1.1)^1 + 55k = \mathbf{\$200,020}$
- **Project Gamma FV:**  $\$50k(1.1)^3 + 30k(1.1)^2 + 25k(1.1)^1 + 25k = \mathbf{\$155,350}$

#### c) Conclusion

**Project Beta** is financially the most viable because it yields the highest Present Value and Future Value.

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### Question 3: Investment with Specific Yearly Payments

**Problem:** Payments: \$150 (Y1-3), \$250 (Y4), \$300 (Y5), \$500 (Y6). Rate = 8%.

- **PV Calculation:**
  - $\$PV \setminus (Y1-3 \setminus \text{Annuity}) = 150 \times \left[ \frac{1 - (1.08)^{-3}}{0.08} \right] = 386.56\$$
  - $\$PV \setminus (Y4) = 250 / (1.08)^4 = 183.76\$$
  - $\$PV \setminus (Y5) = 300 / (1.08)^5 = 204.17\$$

- $SPV(Y6) = 500 / (1.08)^6 = 315.08\$$
- **Total PV = \$1,089.57**
- **FV Calculation (at Year 6):**
  - $FV = PV \times (1.08)^6 = 1,089.57 \times 1.5869 = \mathbf{\$1,729.04\$}$

## Question 4: Comparing Stream A and Stream B

**Problem:** Calculate PV at 5%.

- **Stream A:** \$450 (Y1), \$450 (Y2), \$0 (Y3), \$150 (Y4), \$450 (Y5), \$250 (Y6).
  - $SPV = \frac{450}{1.05} + \frac{450}{1.05^2} + 0 + \frac{150}{1.05^4} + \frac{450}{1.05^5} + \frac{250}{1.05^6} = \mathbf{\$1,399.71\$}$
- **Stream B:** \$0 (Y1), \$250 (Y2), \$450 (Y3), \$450 (Y4), \$450 (Y5), \$150 (Y6).
  - $SPV = 0 + \frac{250}{1.05^2} + \frac{450}{1.05^3} + \frac{450}{1.05^4} + \frac{450}{1.05^5} + \frac{150}{1.05^6} = \mathbf{\$1,450.41\$}$

## Question 5: NFL Contract Negotiation

**Problem:** Opportunity cost = 7%. Compare three 4-year contracts.

- **Contract 1:** \$3M (Y1), \$3M (Y2), \$3M (Y3), \$3M (Y4).
  - $SPV = 3 \times \left[ \frac{1 - (1.07)^{-4}}{0.07} \right] = \mathbf{\$10.16M\$}$
- **Contract 2:** \$2M (Y1), \$3M (Y2), \$4M (Y3), \$5M (Y4).
  - $SPV = \frac{2}{1.07} + \frac{3}{1.07^2} + \frac{4}{1.07^3} + \frac{5}{1.07^4} = \mathbf{\$11.57M\$}$
- **Contract 3:** \$7M (Y1), \$1M (Y2), \$1M (Y3), \$1M (Y4).
  - $SPV = \frac{7}{1.07} + \frac{1}{1.07^2} + \frac{1}{1.07^3} + \frac{1}{1.07^4} = \mathbf{\$8.99M\$}$

**Decision:** The quarterback should choose **Contract 2** as it has the highest Present Value.

